Long Context or Retrieval Augmentation? A Survey of Trade-offs in Large Language Models

Team: 4 tokens | Track A: Survey

Abstract. We propose a focused survey of how large language models use external text for long-document and multi-document question answering. We will organize full-context, retrieval-based, compression-based, and hybrid approaches, and compare their answer quality, resource requirements, and failure modes. The intended outcome is an evidence-based account of when each strategy is appropriate, with explicit attention to the comparability of published evaluations.

Keywords: long-context language models; retrieval-augmented generation; question answering; efficiency; evaluation

1 Background and Motivation

Answering questions over lengthy documents requires a system to identify relevant evidence and combine it into a supported answer. Retrieval-augmented generation (RAG) supplies a language model with selected passages from an external collection [1]. Long-context (LC) models offer another strategy: directly placing a large document or document collection in the prompt. This raises a practical question: when should a system retrieve, read the full context, or combine the two?

Neither strategy removes the evidence-use problem. Retrieval can omit necessary passages or separate related information; full-context input can contain distractions and may be used unevenly. The positional sensitivity documented in *Lost in the Middle* illustrates why a large supported window is not equivalent to effective context use [2]. Direct comparisons have examined quality–cost trade-offs and hybrid routing [3], while Later work shows that simple retrieval baselines preserving document structure can match or outperform more complex pipelines under matched token budgets [4]. Our initial reading of the seed literature suggests a consistent pattern: advantages are driven less by the strategy than by three conditions—whether the corpus fits the window, how the evidence is dispersed, and how strongly passages depend on one another. Decomposition-based long-context methods degrade specifically on documents with strong cross-sentence dependency [5], whereas larger retrieval units target the multi-hop cases where passage-level retrieval fragments evidence [6]. We treat these observations as hypotheses to test across the full corpus; they motivate a survey that explains the conditions behind reported advantages, rather than declaring a universal winner.

2 Scope and Research Questions

Our primary scope is **text-only, document-grounded question answering**, covering single-document, multi-document, and multi-hop questions. We distinguish a supplied document set from retrieval over a larger corpus: a full-context baseline may be feasible in the former but infeasible in the latter. Summarization and synthetic recall tasks serve only as supporting evidence about information coverage and context use. Multimodal tasks, model pre-training, and a comprehensive review of attention architectures are outside the scope.

Here, *LC* denotes direct full-context input when it fits the model's window; *RAG* denotes selecting external evidence before generation. They are not mutually exclusive: a long-context model can also serve as a RAG reader. This proposal addresses three research questions:

- **RQ1. Quality and failure modes:** How do evidence location, dispersion, context dependency, and irrelevant content affect the relative strengths of full-context and retrieval-based approaches?
- **RQ2. Efficiency and comparison:** How do token budgets, retrieval overhead, one-time preprocessing, and repeated-query workloads change the quality–cost–latency trade-off, and under what workload assumptions does each strategy dominate?
- **RQ3. Combination strategies:** How do mechanisms such as larger retrieval units, context compression, and adaptive routing balance local evidence selection with broader context coverage, and what limitations remain? We will also examine other relevant mechanisms identified during the survey.

3 Preliminary Taxonomy and Literature

We classify systems by **how they construct the evidence presented to the answering model**. The categories describe strategies, not exclusive model families; multi-stage systems may receive multiple labels.

1. **Full-context input:** provide the available source text directly, subject to the input budget. Analyze context utilization and positional effects [2].
2. **Retrieval-based selection:** retrieve passages or larger document units, optionally with reranking. RAG supplies the foundation [1]; LongRAG studies larger retrieval units and long-context readers [6].
3. **Compression and structured access:** summarize or organize documents before answering, with access to original evidence where supported. Representative approaches include RAPTOR and ReadAgent [7, 8].
4. **Adaptive and hybrid strategies:** route queries between retrieval and full-context processing, or adjust

the evidence-access strategy. Self-Route and context-dependency-aware routing provide starting points [3, 5].

Across these categories, we annotate each system for evidence granularity, preservation of document order, compression loss, reader model, and routing overhead. LongBench and LongBench v2 provide complementary benchmark perspectives [9, 10]; the controlled baseline study of Laitenberger et al. [4] guides our comparison criteria. Together, the ten cited works form the preliminary reading list, spanning foundational methods, direct comparisons, hybrid systems, and evaluation.

4 Survey Method and Comparison Plan

Literature collection. The search covers the ACL Anthology, major conference proceedings, and arXiv using combinations of "long context," "retrieval-augmented generation," "document question answering," "context compression," and "hybrid routing." Starting from the ten seed papers, we follow their references and subsequent citations.

Selection and extraction. We include work that studies evidence access for document-grounded QA, directly compares LC and retrieval, or supplies a relevant method or benchmark, and exclude unsupported product claims, duplicate versions, and work outside the scope. A second member checks the records used for central claims; unresolved disagreements are documented.

5 Expected Outcomes and Feasibility

The project will deliver an 8-page survey using the official ACL LaTeX template, excluding references and appendices, with at least 20 relevant academic references. The survey will present a technical taxonomy, a method-comparison matrix, and an analysis of the strengths, limitations, and applicable scenarios of full-context, retrieval-based, and hybrid approaches. It will synthesize evidence on their quality–efficiency trade-offs, identify gaps in existing research, and discuss future directions. The team will also prepare the required presentation slides and individual contribution statement.

The project is feasible because representative methods, comparative studies, and relevant benchmarks provide a foundation for the review. Restricting the scope to text-only, document-grounded question answering keeps the workload manageable. A shared literature extraction framework will support consistent comparisons and help distinguish methodological differences from differences in evaluation settings.

6 Project Timeline

By October 12: Confirm scope and team allocation; submit the proposal with seed references and an initial taxonomy.

First two weeks after proposal: Read anchor papers, collect at least 20 candidate papers, and populate the shared extraction sheet.

By the midterm milestone (TBA): Complete the main collection, refine the taxonomy, and draft partial comparisons; submit the 2–3-page report and at most five progress slides.

After midterm: Resolve gaps and conflicting findings, update the search, and draft and cross-review the final survey.

Before presentation/final submission (TBA): Finalize the report, references, and contribution statement; rehearse the 8-minute talk and 3-minute Q&A.

7 Team Members and Responsibilities

Member	Student ID	Primary Responsibility
Aozhe Zhang	2313447	Scope and research questions; direct-comparison literature; synthesis and report integration.
Junhui Yu	2313486	Full-context methods and failure analysis; draft the corresponding comparison section.
Qianxia Jing	2312130	Retrieval, compression, and hybrid methods; maintain the taxonomy.
Zhibo Yao	2313557	Benchmarks and cost accounting; maintain the evidence matrix and presentation outline.

Table 1. Team members and their primary responsibilities.

All members will read shared anchor papers, cross-check another member's work, contribute to writing and presentation/Q&A, and maintain a contribution log.

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